

1) Given:

$$G_x = 24$$

$$G_y = 18$$

Gradient Magnitude

$$G = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{24^2 + 18^2}$$

$$= \sqrt{576 + 324}$$

$$= \sqrt{900}$$

$$= 30$$

Gradient Magnitude = 30

Gradient Direction

$$\theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

$$\theta = \tan^{-1} \left(\frac{18}{24} \right)$$

$$\theta = \tan^{-1} (0.75) \approx 36.87^\circ$$

Gradient Direction $\approx 36.87^\circ$

2)

Give gradient magnitudes:

Pixel P_1 P_2 P_3 P_4 P_5

Magnitude 25 50 90 60 30

Since the gradient direction is horizontal, each pixel is compared with its left and right neighbours.

$P_1 = 25 \rightarrow$ boundary pixel $\rightarrow$ retained
 $P_2 = 50 \rightarrow$ less than $P_3 (90) \rightarrow$ suppressed
 $P_3 = 90 \rightarrow$ greater than $P_2 (50)$ and $P_4 (60) \rightarrow$ retained
 $P_4 = 60 \rightarrow$ less than $P_3 (90) \rightarrow$ suppressed
 $P_5 = 30 \rightarrow$ boundary pixel $\rightarrow$ retained

Result after NMS:

	P_1	P_2	P_3	P_4	P_5
Pixel					
NMS Result	25	0	90	0	30

Remaining edge pixels: P_1, P_3 and P_5

3) Double Thresholding

Given:

High Threshold = 100
 Low Threshold = 50

Rules:

magnitude $\geq 100 \rightarrow$ Strong Edge
 $50 \leq$ magnitude $< 100 \rightarrow$ Weak Edge

Pixel	Magnitude	classification
P_1	135	Strong Edge
P_2	90	Weak edge
P_3	45	Non-edge
P_4	120	strong edge
P_5	70	Weak Edge
P_6	20	Non Edge

Answers:
strong edges : P_1, P_4
weak edges : P_2, P_5
Non-edges : P_3, P_6

4)

pixel	P_1	P_2	P_3	P_4	P_5	P_6	P_7
Magnitude	130	95	40	115	65	25	10

Thresholds:

- High = 100
- Low = 50

1. strong edges

Magnitude ≥ 100 :

P_1, P_4, P_7

2. weak edges

$50 \leq \text{Magnitude} < 100$:

P_2, P_5

3. Non-edges

Magnitude < 50 :

P_3, P_6

4. Final edge pixels after Hysteresis

• P_2 is connected to strong edge P_1 , so

P_2 is retained

• P_5 is Not connected to any strong edge,

So P_5 is removed.

Final edge pixels - P_1, P_2, P_4, P_7

Final Answer Table

Pixel	classification	After Hysteresis
P_1	Strong	Edge
P_2	Weak	Edge
P_3	Non-edge	Removed
P_4	Strong	Edge
P_5	Weak	Removed
P_6	Non-edge	Removed
P_7	Strong	Edge

Final edges: P_1, P_2, P_4, P_7

5) Hough Circle Transform:

$$\rho = \sqrt{(x-a)^2 + (y-b)^2}$$

i) edge points:-

$$P_1 = (5, 4)$$

$$P_2 = (7, 6)$$

$$C_1 = (2, 4)$$

$$C_2 = (4, 3)$$

$$\text{For } C_1 = (2, 4)$$

$$P_1 \text{ and } C_1: \rho = \sqrt{(5-2)^2 + (4-4)^2} \\ = \sqrt{3^2 + 0^2}$$

$$\boxed{r=3}$$

$$P_2 \text{ and } C_2 : r = \sqrt{(7-5)^2 + (6-4)^2}$$

$$= \sqrt{25+4}$$

$$= \sqrt{29}$$

$$r = 5.39$$

$$P_1 \text{ and } C_2 : r = \sqrt{(5-4)^2 + (4-3)^2}$$

$$= \sqrt{1+1}$$

$$= \sqrt{2}$$

$$\boxed{r=1.41}$$

$$P_2 \text{ and } C_2 : r = \sqrt{(7-4)^2 + (6-3)^2}$$

$$= \sqrt{9+9}$$

$$= \sqrt{18}$$

$$\boxed{r = 4.24}$$

6) Calculate the radius:

Given:

$$r = (8, 6)$$

$$c = (4, 3)$$

$$r = \sqrt{(x-a)^2 + (y-b)^2}$$

$$= \sqrt{(8-4)^2 + (6-3)^2} = \sqrt{25}$$

$$\boxed{r=5}$$

7)

Calculate the radius

$$C = (5, 4)$$

$$P = (8, 8)$$

$$r = \sqrt{(8-5)^2 + (8-4)^2}$$

$$= \sqrt{3^2 + 4^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$\boxed{r = 5}$$

